1. What the model actually receives

You asked what the detector is being handed, so this starts there and works backwards to the optics. Short version: the numbers in `sizing-calculations.md` §8 were computed for a different sensor than the one we are buying, and the difference matters more to you than to anyone else on the team.

We are buying an **Arducam IMX477** with a 6 mm S-mount lens. Sony specifies it as type 1/2.3: 4056×3040 on a **1.55 µm** pitch. The sizing chapter assumes "1/1.8 in… **1.82 µm** pitch". Same pixel *count*, different pixel *size* — so resolution on target is not what the chapter says.

Here is the target through the pipeline, at the 40 m search altitude, taking a supine adult as 1.7 m by 0.5 m:

	GSD (cm/px)	Target (px)	Area (px^2)	Of a 640 tile	COCO class
Sensor, native	1.03	165×48	7960	26 %	medium
After $2\times$ downsample	2.07	82×24	1990	13 %	medium
Chapter figure, 60 m, $2\times$	3.64	47×14	642	7 %	**small**

The last row is the one to look at. The 47 px figure currently in the proposal puts the target at 642 px^2, which is *below* COCO's small-object threshold of 1024 px^2. On the sensor we are actually buying, at 40 m, it sits at 1990 px^2 — comfortably in the *medium* band.

That is not a cosmetic difference. Any published recall figure you benchmark against reports AP by that size class, and small-object AP is routinely a large fraction below medium on the same model. If we quote the 47 px number we are implicitly promising small-object performance we do not have to accept.

2. Where those numbers come from

Everything follows from three quantities: pixel count, pixel pitch, focal length. Active area is count times pitch,

$$w = 4056 \times 1.55\,\mu\text{m} = 6.287\,\text{mm}, \qquad h = 4.712\,\text{mm}, \qquad d = 7.86\,\text{mm},$$

which is the type 1/2.3 format Sony quotes. Half-fields are arctangents of half the sensor over f, exactly, with no small-angle approximation:

$$\theta = 2\arctan\left(\frac{s}{2f}\right) \implies \text{HFOV} = 55.3^\circ, \quad \text{VFOV} = 42.9^\circ, \quad \text{DFOV} = 66.4^\circ.$$

Sensor and ground are similar triangles about the lens, so ground sample distance is just

$$\text{GSD} = \frac{pH}{f} = \frac{1.55\,\mu\text{m} \times 40\,\text{m}}{6\,\text{mm}} = 1.03\,\text{cm/px}.$$

One pixel subtends 14.80 mdeg, which is what sets the blur tolerance in §4.

AGL (m)	**GSD** (cm/px)	**Target** (px)	**Footprint** (m)	**Frames/target**
30	0.78	219	31.4×23.6	6
40	1.03	165	41.9×31.4	8
60	1.55	110	62.9×47.1	12

3. Tiling, and one property worth checking

Tile count depends on pixel count, which did not change, so the inference budget is untouched: at $2\times$ downsample and 2 Hz we still get 12 tiles per frame and 24 inferences per second.

Downsample	**Frame**	**Tiles**	**Inferences/s**	**GSD** (cm/px)	**Target** (px)
$1\times$	4056×3040	48	240	1.03	165
$2\times$	2028×1520	12	24	2.07	82

The property worth checking is the overlap. At 640 px tiles and 20 % overlap the shared margin is 128 px, and the target at 40 m is 82 px along its long axis. **The overlap is wider than the target**, so no survivor can be cut by a tile boundary without appearing whole in the neighbouring tile. That holds at every altitude in the table above, and it is the reason the overlap is 20 % rather than something smaller.

4. Blur, which turns out not to bind

Forward motion smears the image by $v t_{\text{exp}}$ on the ground, so holding smear under one pixel means $t_{\text{exp}} \leq \text{GSD}/v$. At 8 m/s:

AGL (m)	**Longest exposure** (ms)	**Shutter**
30	0.97	1/1032
40	1.29	1/774
60	1.94	1/516

`SYS-45` already mandates $1/1000$ s or faster, which clears every altitude. In Indian daylight that pushes ISO down rather than exposure up, so it costs nothing in noise either. Blur is not what will limit recall; target size is.

5. Sampling in time: how many looks the fusion gets

This is the section I would read first, because it does not currently close.

A target sits in frame for as long as the along-track footprint takes to pass over it. The along-track footprint is the vertical field of view projected on the ground, so

$$D = 2H \tan \frac{\text{VFOV}}{2}, \qquad t_{\text{dwell}} = \frac{D}{v}, \qquad n_{\text{looks}} = t_{\text{dwell}}\, r,$$

for capture rate r and ground speed $v = 8$ m/s. Forward overlap between consecutive frames is $1 - (v/r)/D$.

AGL (m)	**Along-track** (m)	**Dwell** (s)	**Rate** (Hz)	**Looks**	**Fwd overlap** (%)
30	23.6	2.94	2	**5.9**	83
30	23.6	2.94	3	**8.8**	89
30	23.6	2.94	5	14.7	93
40	31.4	3.93	2	**7.9**	87
40	31.4	3.93	3	**11.8**	92
40	31.4	3.93	5	19.6	95
60	47.1	5.89	2	**11.8**	92
60	47.1	5.89	3	17.7	94
60	47.1	5.89	5	29.4	97

`SYS-46` requires at least 12 looks per target per pass, so that multi-frame fusion has enough independent observations to converge. **At 40 m and 2 Hz we get 7.9.** That is a 35 % shortfall, and it is not a rounding argument.

The figure quoted elsewhere — about 14 looks at 2 Hz — is correct for 60 m on the wider sensor the sizing chapter assumed. On the sensor we are buying, at the altitude the simulations fly, it does not hold. Meeting 12 looks at 40 m needs

$$r \geq \frac{n\,v}{D} = \frac{12 \times 8}{31.4} = 3.06\,\text{Hz},$$

which at 12 tiles per frame is 37 inferences per second rather than 24. That is the trade in front of us: raise the capture rate and pay about half again in inference load, fly slower, fly higher, or relax the look count. I do not think it should be decided by whoever notices it last.

6. Focus and rolling shutter

Two things worth ruling in or out before they surprise us.

Focus is not a concern. The hyperfocal distance for this lens is

$$H_{\text{hyp}} = \frac{f^2}{N\,c} + f = \frac{(6\,\text{mm})^2}{2.8 \times 0.0031\,\text{mm}} + 6\,\text{mm} = 4.15\,\text{m},$$

taking the circle of confusion as 2 pixels rather than a film-era constant, since the sensor is what resolves. Focused at that distance everything from about 2.1 m to infinity is acceptably sharp, and we operate at 30–60 m. Nothing to do.

Rolling shutter is small but not zero. The IMX477 reads out row by row over roughly 25 ms at full frame, during which the aircraft moves $v\,t_{\text{ro}} = 0.20\,\text{m}$. Top to bottom that is **19 px** of skew at 40 m.

For detection this is negligible: a target spans only part of the frame height, so the skew *within* one target is about 1.0 px — less than the blur budget. For geolocation it is not negligible, because a detection's row position now carries up to 20 cm of along-track bias depending on where in the frame it landed. If the geotag pipeline does not already correct for readout row, that is a systematic error sitting inside our budget rather than a random one.

7. Imagery volume

Relevant because we still need training data, and because the storage line in the parts list has to hold a mission.

One frame is 4056 × 3040 at 12-bit packed, so 18 MB raw or about 2 MB compressed. A single 196 s sweep at 2 Hz is 392 frames: **7.2 GB raw, 0.8 GB compressed** per aircraft, per sweep. Three aircraft over a full mission with retries will comfortably fill the 256 GB module if we record raw, and will not come close if we record JPEG.

That is a decision worth making deliberately rather than discovering on the day: raw gives you the data to retrain against, JPEG gives you the endurance to keep recording. If we want a dataset out of the competition itself, raw on at least one aircraft is the way to get it.

8. What it costs the flight side

For completeness, since the same substitution moves the coverage numbers in the opposite direction. A smaller pixel behind the same lens narrows the field and finens the GSD in the same proportion, so at 40 m:

At 40 m AGL	**IMX477**	**Sizing chapter**	**Change, %**
Horizontal field of view, °	55.30	63.20	-12.5
Ground sample distance, cm/px	1.03	1.21	-14.8
Survivor along the long axis, px	164.52	140.11	+17.4
Swath width, m	41.91	49.21	-14.8
Line spacing, m	29.34	34.45	-14.8
Transects per drone	7.00	6.00	+16.7
Sweep time per drone, s	195.75	166.93	+17.3

We gain 17 % more pixels on target and pay 17 % more sweep time — seven transects per drone instead of six. Against 250 marks for detection and an 1800 s budget I think that favours the sensor we are buying, but it was never a decision anyone made.

9. Where to start

- **Train across scales, then fine-tune towards 82 px.** That is the target size at 40 m after downsampling, but it is a centre rather than a fixed point — ground sample distance is proportional to altitude, so any altitude error moves it. A ± 5 m band about 40 m puts the same person anywhere between 73 and 94 px, and terrain relief widens that further. A model trained only at 82 px would be brittle to exactly the variation the aircraft guarantees, so keep normal scale augmentation and bias the fine-tuning towards the operating point.
- **Check the size band when you read a published number.** Detection papers report average precision separately for small, medium and large targets, using the pixel-area thresholds in §1. A headline figure quoted without its band is not comparable to anything we are doing — ours sits in *medium*, and small-object numbers are usually much worse.
- **HERIDAL and SARD are the obvious starting datasets.** When you benchmark on either, record which band the result came from, so we can tell whether it transfers.
- **The measurement that unblocks a real decision is recall against target size.** If you can say what recall looks like at 82 px versus 47 px, that settles an argument nobody can currently settle: the design point says survey at 60 m, every simulation we have flown uses 40 m, and the sizing chapter proposes 40 m pending exactly this measurement.
- **It does not have to be precise to be useful.** Even a rough curve of recall against pixel height, on whatever data you have, decides the altitude and lets the documents be corrected once instead of twice.